SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	G. Akshya	**Reg. No.:**	192425132
Section / Batch:	Slot A/2024	**Date:**	14/10/2026

Code of Conduct

I G. Akshya 192425132 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *G. Akshya*

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

CO 1 Assessment Tool-3

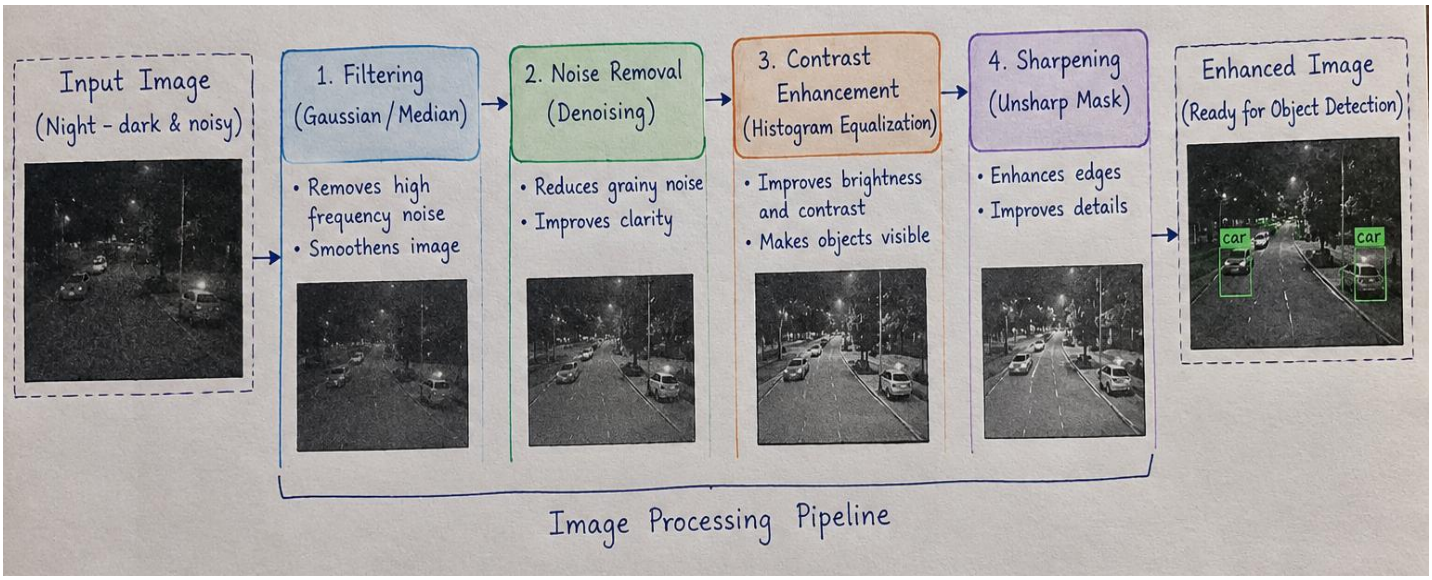
Q1. Image Enhancement for Night-Time CCTV Images

Introduction

CCTV cameras may capture dark, low-contrast, and noisy images at night because of limited illumination and sensor noise. These problems can hide facial details, clothing, and object boundaries. Image preprocessing improves visibility before object detection, but it cannot recover details that were never captured.

Techniques Used

- **Filtering:** A median filter removes salt-and-pepper noise while preserving many edges. A Gaussian filter reduces random high-frequency noise, although excessive smoothing can blur small details.
- **Noise removal:** Apply a suitable denoising filter before strong enhancement. The filter strength should be selected carefully so that people, vehicle edges, and other useful details are not erased.
- **Contrast enhancement:** Histogram equalization spreads intensity values across the available range. CLAHE enhances local contrast in small image regions and can reveal details in shadows while limiting excessive amplification.
- **Sharpening:** Unsharp masking or a high-pass sharpening method strengthens edges after denoising and contrast adjustment. Over-sharpening should be avoided because it can amplify noise and create halos.



Recommended Processing Sequence

Input CCTV frame → convert to grayscale when color is not required → denoise with a median or Gaussian-based method → improve local contrast using CLAHE → sharpen important edges → normalize/resize the image → send the enhanced frame to the object detector → review detection confidence.

Evaluation and Limitations

Compare the original and processed frames using visibility of people and objects, edge clarity, noise level, and object-detection precision/recall. Enhancement should be tested on different cameras and lighting conditions. Brightness enhancement may also amplify noise, and no filter can fully restore severely blurred or completely underexposed details.

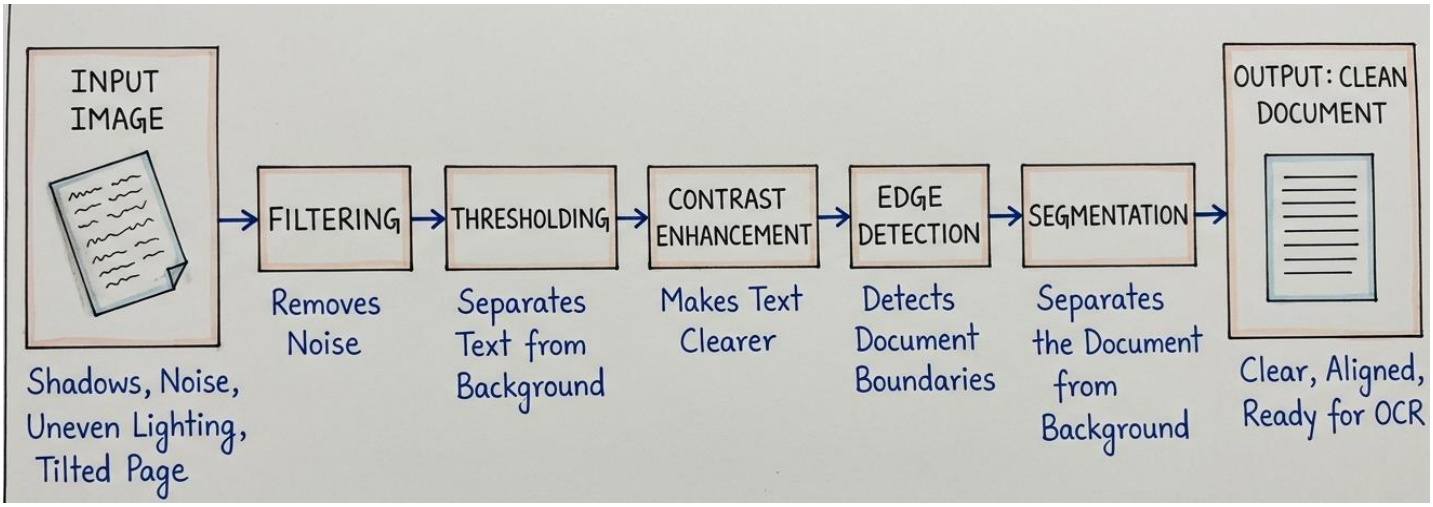
Result

A carefully ordered combination of denoising, contrast enhancement, and sharpening can make night-time CCTV images more suitable for object detection while reducing the risk of losing important edges.

Q2. Image Processing for an Automated Document-Scanning System

Introduction

Scanned documents can contain shadows, uneven illumination, skew, noise, and blurred characters. The goal is to produce a clean, aligned document image so that text segmentation and optical character recognition (OCR) can work more reliably.



Techniques and Their Roles

- **Filtering:** Median filtering can remove isolated specks; Gaussian filtering can reduce mild noise. Use small kernels to avoid damaging thin strokes in letters.
- **Thresholding:** Global thresholding assigns pixels to foreground or background using one threshold and works well with uniform lighting. Adaptive thresholding calculates local thresholds and is more suitable for shadows and uneven illumination.
- **Contrast enhancement:** Normalize intensity or use local contrast enhancement to make faint text more visible. Avoid making paper texture or shadows too prominent.
- **Edge detection:** Sobel or Canny edge detection can help identify page boundaries and text structures. Edge maps are useful supporting information, but should not replace the document image.

- **Segmentation:** Separate the document from the background, then identify text lines, words, or characters where required. Connected-component analysis can help isolate text regions.

Recommended Processing Pipeline

Capture image → detect page boundary → correct perspective and deskew → denoise → correct uneven illumination/enhance contrast → apply adaptive or global thresholding → remove small unwanted components → segment text regions → pass the result to OCR.

Practical Considerations

A tilted page should be deskewed before line segmentation. Adaptive thresholding is generally more robust when illumination varies, while global thresholding is simpler for uniformly lit pages. Very aggressive thresholding can break thin characters or merge nearby letters. Quality can be checked by visual inspection and OCR character/word accuracy.

Result

Combining illumination correction, filtering, thresholding, edge-based boundary detection, and segmentation creates cleaner and better-aligned document images, improving the reliability of downstream OCR.

Q3. Segmentation Methods for Detecting Damaged Apples

Introduction

A fruit-processing system must distinguish apples from the conveyor background and identify damaged areas despite shadows, reflections, different backgrounds, and brightness changes. Segmentation isolates the fruit or candidate defect regions for later analysis.

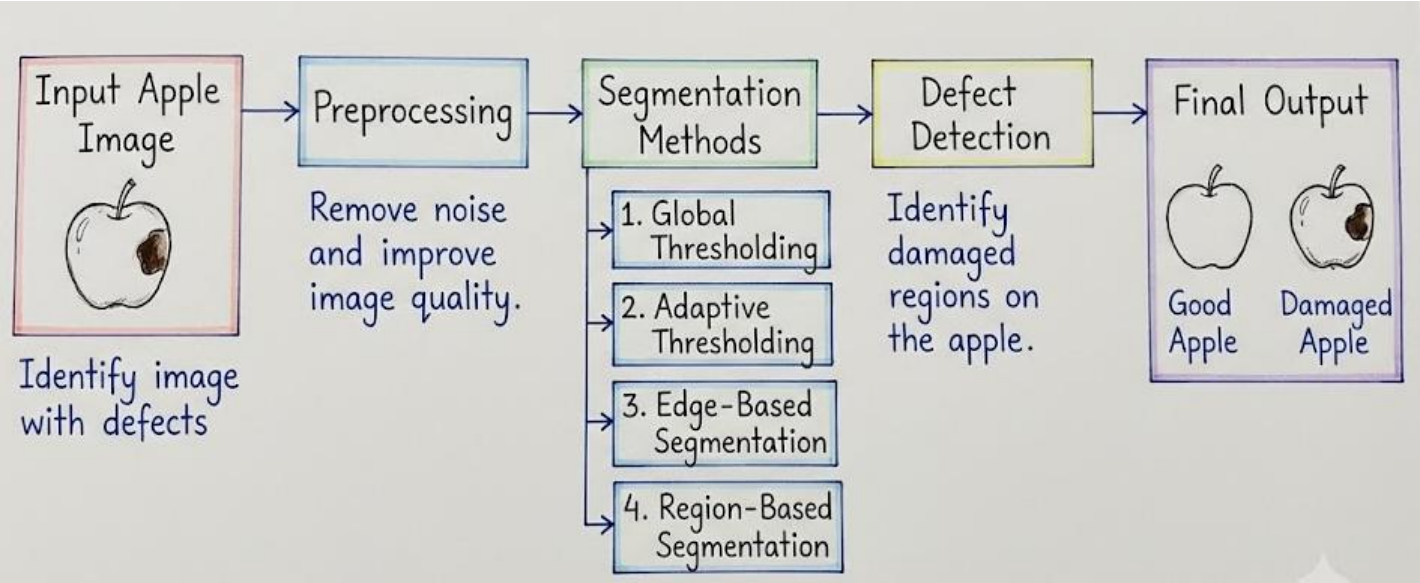
Comparison of Segmentation Techniques

- **Global thresholding:** Uses one intensity threshold for the entire image. It is fast and simple when lighting and background are consistent, but performs poorly under shadows or brightness variations.
- **Adaptive thresholding:** Uses local thresholds, making it more tolerant of uneven illumination. It may still struggle with shiny reflections, textured skin, or weak boundaries.
- **Edge-based segmentation:** Uses operators such as Canny to find sharp intensity changes around the apple and around possible defects. It can preserve shape boundaries but may produce broken edges or false edges from reflections.
- **Region-based segmentation:** Groups neighboring pixels with similar color, intensity, or texture. It can isolate coherent fruit areas, but results depend on the similarity rule and may merge defects with healthy skin.

Recommended Combined Method

Capture image under controlled lighting → correct color/brightness and denoise → segment the apple from the conveyor using color information and adaptive thresholding or a suitable region method → refine the apple

boundary with edge information → inspect the fruit surface using color, texture, and local intensity differences → classify candidate regions as healthy surface, defect, or reflection → flag apples whose defect score exceeds a validated threshold.



Handling Real-World Variation

Use diffuse lighting and a plain conveyor background where possible. Normalize brightness, avoid saturated highlights, and validate the method on apples of different colors, sizes, orientations, and defect types. Reflections should not automatically be treated as damage; combining texture and color evidence can reduce false alarms.

Evaluation and Result

Evaluate segmentation with intersection-over-union or boundary accuracy, and defect detection with precision, recall, and false-positive rate. No single segmentation technique is best in every scene; a combination of adaptive thresholding, region grouping, and edge refinement is more robust than relying only on a global threshold.

Q4. Lane Detection Using Image-Processing Techniques

Introduction

A vehicle camera must identify lane markings despite shadows, rain, faded paint, road texture, and other vehicles. The processing pipeline should reduce irrelevant variation while preserving the line-like structures that indicate lane boundaries.

Processing Stages and Challenges

- **Grayscale conversion:** Simplifies intensity processing and reduces color-channel complexity. However, yellow and white lane markings may become similar to the road in grayscale.

- **Filtering:** Gaussian filtering reduces small texture variations and sensor noise. Too much smoothing can remove thin or damaged lane markings.
- **Contrast enhancement:** Local contrast enhancement can make faded lane lines more visible. It may also increase shadows, wet-road reflections, or headlight glare.
- **Edge detection:** Canny edge detection identifies strong boundaries that may correspond to lane markings. Road cracks, vehicle edges, and shadows can create false edges.
- **Thresholding:** Intensity or color-informed thresholding isolates likely lane pixels. Fixed thresholds may fail when sunlight, tunnels, rain, or road surfaces change.
- **Region of interest (ROI):** Restrict processing to the road area, commonly a trapezoid in front of the vehicle. This reduces irrelevant detections but may miss lanes on sharp curves, slopes, or unusual camera angles.

Recommended Processing Pipeline

Camera frame → grayscale or color-channel selection → mild Gaussian filtering → contrast normalization → edge detection and/or lane-color thresholding → apply road ROI mask → group lane candidates → fit lane curves/lines → track lane positions across consecutive frames → estimate lane boundaries and confidence.

Robustness and Evaluation

Test during daytime, night-time, rain, shadow, glare, worn markings, curves, and partial occlusion. Temporal tracking can reduce flicker, but the system must lower confidence when markings are missing or ambiguous. Evaluate lane-boundary error, detection rate, false detections, and performance across weather conditions.

Result

A staged pipeline combining filtering, contrast improvement, edge detection, thresholding, and an ROI can reduce irrelevant features and improve lane detection, provided each stage is tuned and tested for changing road conditions.

Q5. Image-Processing Pipeline for Waste Classification

Introduction

A municipal waste-sorting system must classify objects such as plastic, paper, and metal from conveyor-belt images. Overlapping objects, shadows, reflections, noise, and different orientations can make classification difficult. A complete pipeline should isolate objects and extract useful visual features before classification.

Complete Pipeline

- **Image acquisition:** Capture images with stable camera position and sufficient resolution. Use controlled, diffuse lighting where practical.
- **Preprocessing:** Resize consistently, correct color/brightness, and remove noise using an appropriate filter. Preserve edges and texture details.

- **Enhancement:** Improve contrast or normalize illumination to reduce the effect of shadows. Avoid excessive enhancement that changes material appearance.
- **Segmentation:** Separate objects from the conveyor using background subtraction, thresholding, colour cues, or edge/region methods. Connected components can identify separate object candidates.
- **Overlap handling:** Use contour analysis, distance-transform/watershed separation, or instance-segmentation methods when objects touch. If objects are heavily occluded, capture additional views or use mechanical spacing.
- **Feature extraction:** Compute color statistics, shape descriptors, size, edge/texture features, and reflectance-related cues. Features should be normalized and robust to rotation and scale.
- **Classification:** Use a trained classifier such as a support-vector machine or a decision-tree-based model with the extracted features, or a suitable image classifier when sufficient labelled training data are available. Train and test using representative examples.
- **Decision and control:** Output the predicted class and confidence, then direct the item to the appropriate sorting mechanism. Low-confidence predictions should be flagged for a second pass or manual review.

Handling Real-World Variations

Use varied training images covering object orientation, color, crushed packaging, dirty materials, shadows, and reflections. Augmentation such as rotation and brightness changes can improve robustness. Metal may create strong highlights, while paper and plastic can have similar colors; therefore, classification should combine multiple features rather than depend on color alone. Overlapping objects remain a major limitation if segmentation cannot separate them.

Evaluation

Measure per-class precision, recall, F1-score, overall accuracy, confusion matrix, and processing time.

Also evaluate object-separation quality and end-to-end sorting accuracy, because a correct classifier cannot compensate for badly segmented objects.

Result

A complete pipeline of preprocessing, enhancement, segmentation, feature extraction, and classification can support reliable waste sorting. Representative data, careful handling of overlaps, and confidence-based decisions are essential for real-world performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____